Breast Cancer Classification using ANN

B.Tech AI Mini Project

Student Name

Department of Artificial Intelligence

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Outline

Problem Statement

- Classify breast tumors as **Malignant (0)** or **Benign (1)**
- Use a simple Artificial Neural Network (ANN)
- Dataset: Breast Cancer Wisconsin (569 samples, 30 features)
- Tools: Python, TensorFlow/Keras, scikit-learn

Dataset Overview

Dataset Information

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Breast Cancer Wisconsin Dataset Information
================================================
Total samples            : 569
Number of features       : 30
Feature type             : Numeric tumor measurements
Classes                  : 0 = Malignant, 1 = Benign
Class counts (full)      : Malignant=212, Benign=357

Training samples         : 455
Test samples             : 114
Train labels shape       : (455,)
Test labels shape        : (114,)

Preprocessing            : StandardScaler (fit on train only)
Train/Test split         : 80% / 20% (stratified)
TensorFlow version       : 2.21.0
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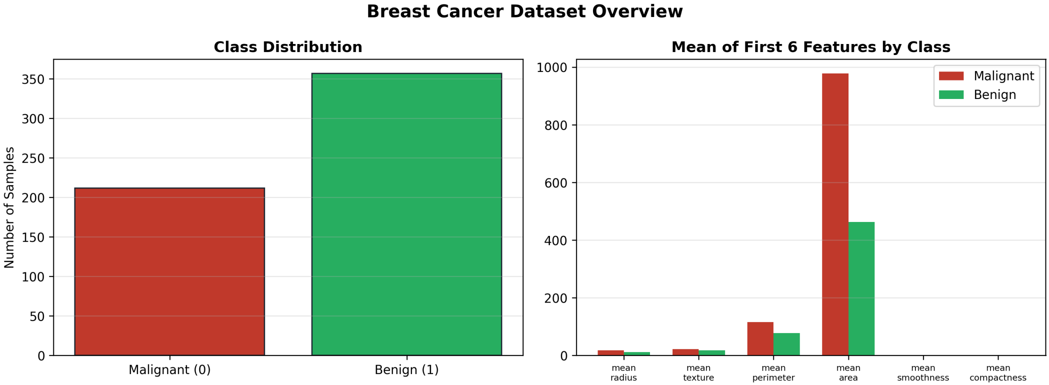
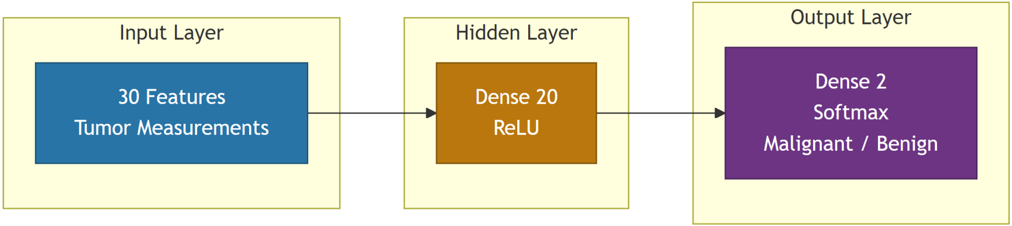
Class Distribution



ANN Architecture



Model Summary

Model Summary

Model: "breast_cancer_ann"

Layer (type)	Output Shape	Param #
hidden_layer (Dense)	(None, 20)	620
output_layer (Dense)	(None, 2)	42

Total params: 662 (2.59 KB)
Trainable params: 662 (2.59 KB)
Non-trainable params: 0 (0.00 B)

Training Progress

Training Progress (10 Epochs)

Epoch	Train Acc	Val Acc	Train Loss	Val Loss
1	52.81%	65.22%	0.7284	0.5902
2	78.97%	86.96%	0.4975	0.4157
3	89.49%	91.30%	0.3653	0.3295
4	92.18%	93.48%	0.2903	0.2803
5	93.64%	93.48%	0.2438	0.2480
6	93.89%	93.48%	0.2121	0.2244
7	94.62%	93.48%	0.1887	0.2055
8	95.11%	93.48%	0.1703	0.1897
9	95.60%	93.48%	0.1555	0.1760
10	96.33%	95.65%	0.1431	0.1639

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Final Training Accuracy   : 96.33%
Final Validation Accuracy : 95.65%
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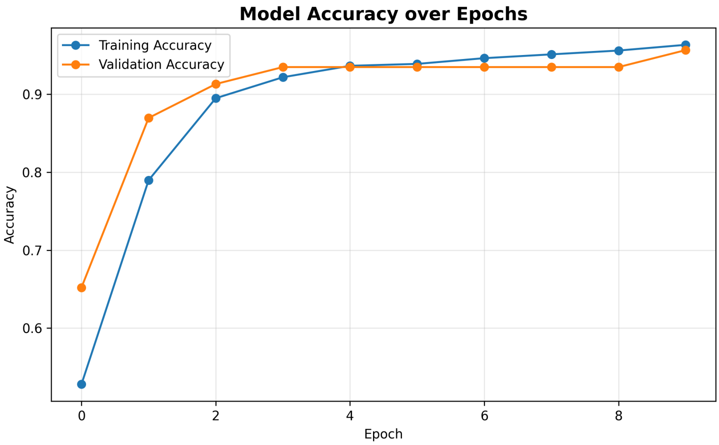
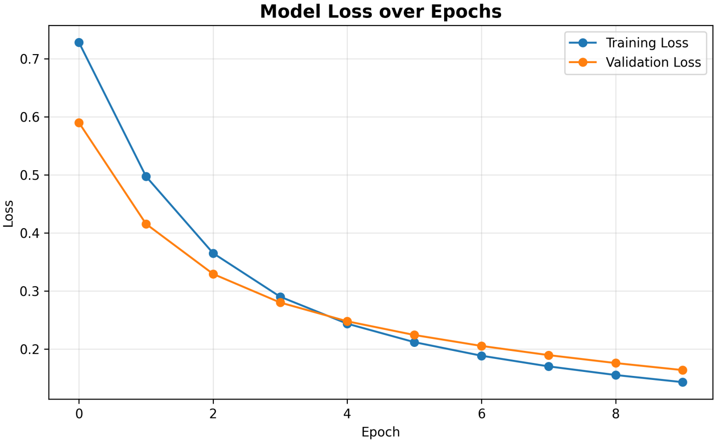
Accuracy Curve



Loss Curve



Model Loss over Epochs
Training Loss
Validation Loss
Loss
Epoch
0.7
0.6
0.5
0.4
0.3
0.2
0
2
4
6
8

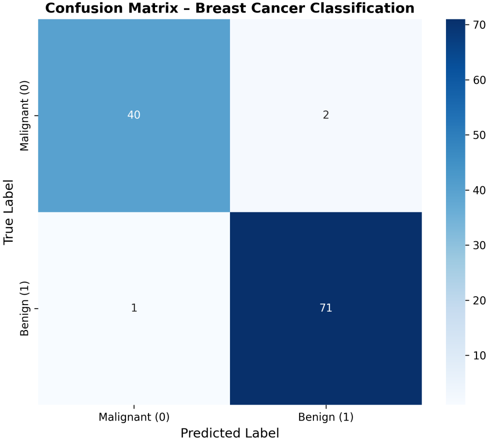
Test Accuracy

Test Accuracy

```
Test Set Evaluation
========================================
Test Loss     : 0.1077
Test Accuracy : 97.37%

Correct predictions : 111 / 114
Misclassifications  : 3
```

Confusion Matrix

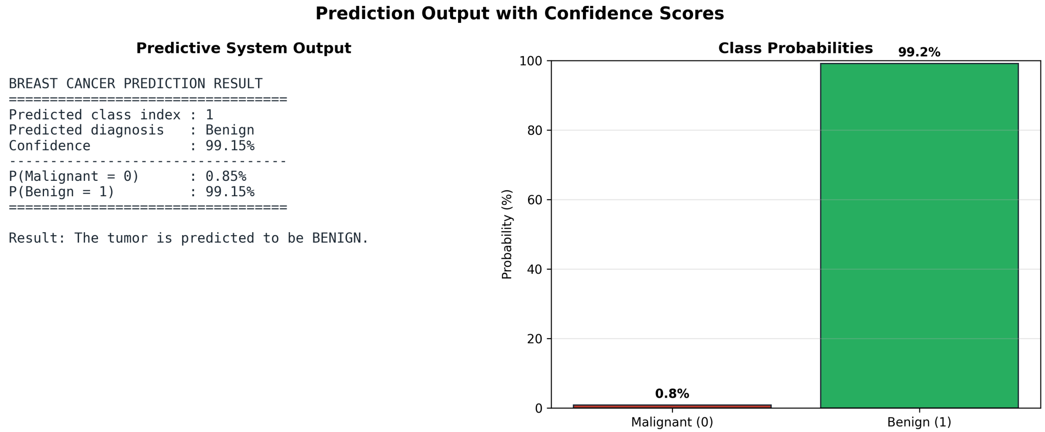


Classification Report

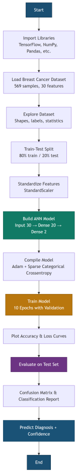
Classification Report

	precision	recall	f1-score	support
Malignant (0)	0.9756	0.9524	0.9639	42
Benign (1)	0.9726	0.9861	0.9793	72
accuracy			0.9737	114
macro avg	0.9741	0.9692	0.9716	114
weighted avg	0.9737	0.9737	0.9736	114

Prediction Demo



Project Workflow



Conclusion

- Built a compact ANN for binary breast cancer classification
- Applied stratified split and feature standardization
- Evaluated with accuracy, confusion matrix, and classification report
- Demonstrated a confidence-aware predictive system

Educational demo only — not a medical diagnostic tool.

Thank You

Questions?